# Current status of aging and social issues Case study in Japan

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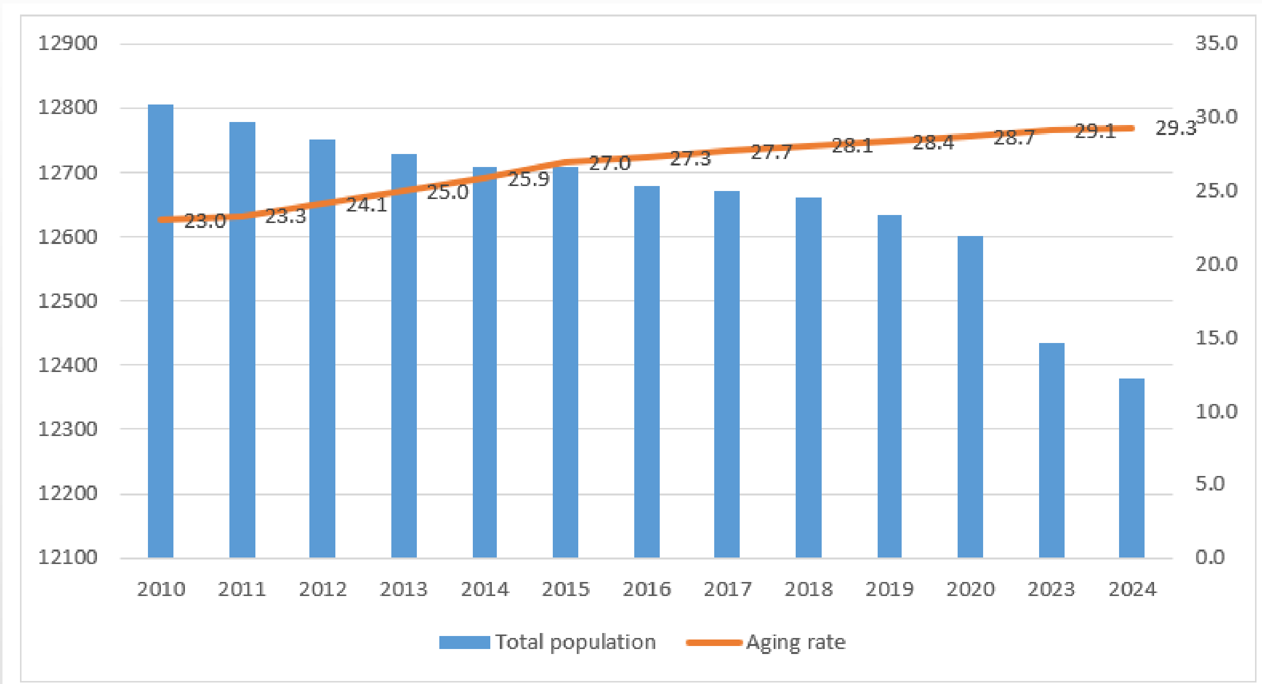
# Current Situation of Aging Society in Japan: Changes in Aging Rate

(Ten thousand people) (%)

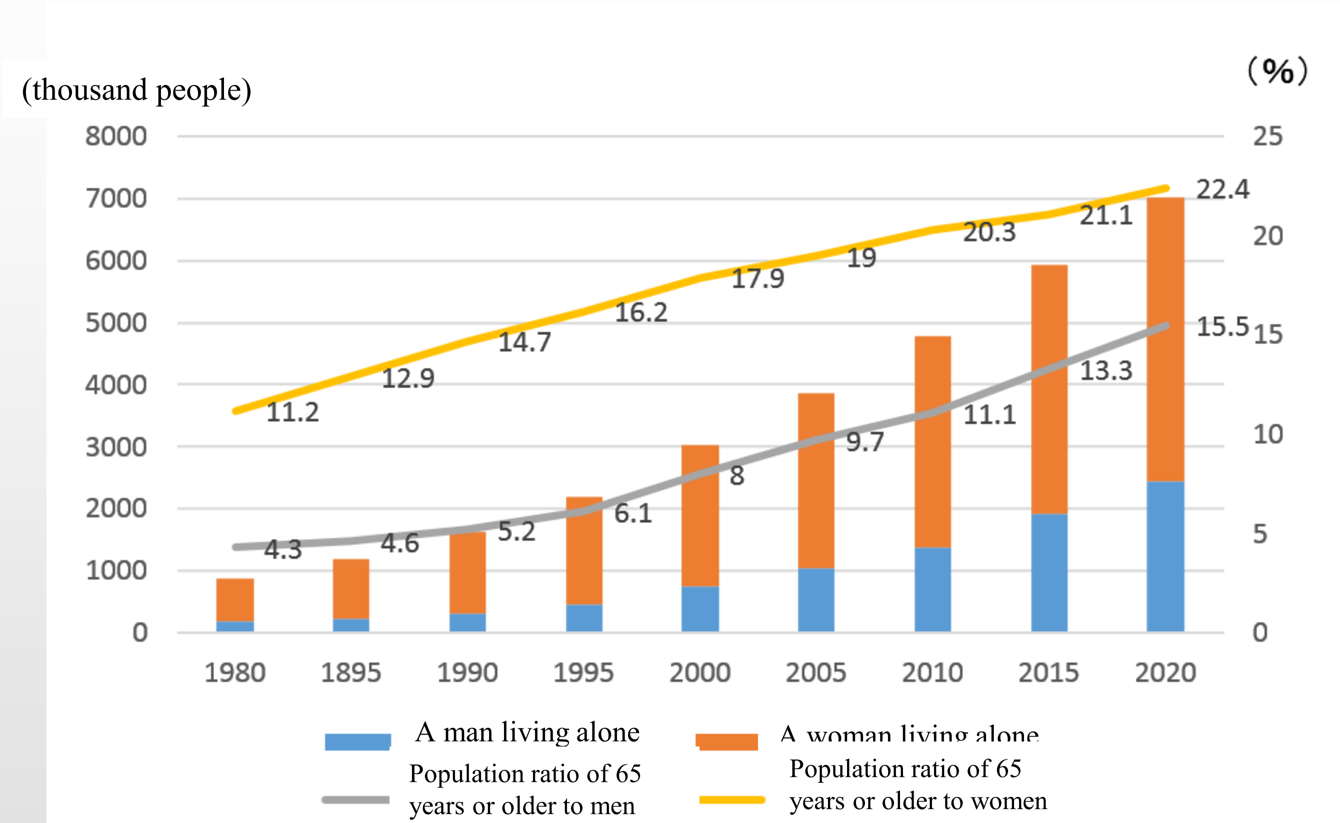



（From 2024Ministry of Health, Labor and Welfare statistical data）

**Figure 1 Changes in the aging rate in Japan**

Japan's population is declining, reaching 123.80 million people in 2024.
On the other hand, the elderly rate is on the rise, reaching 29.3% in 2024.

# Current status of aging society in Japan: Changes in the number of elderly people living alone



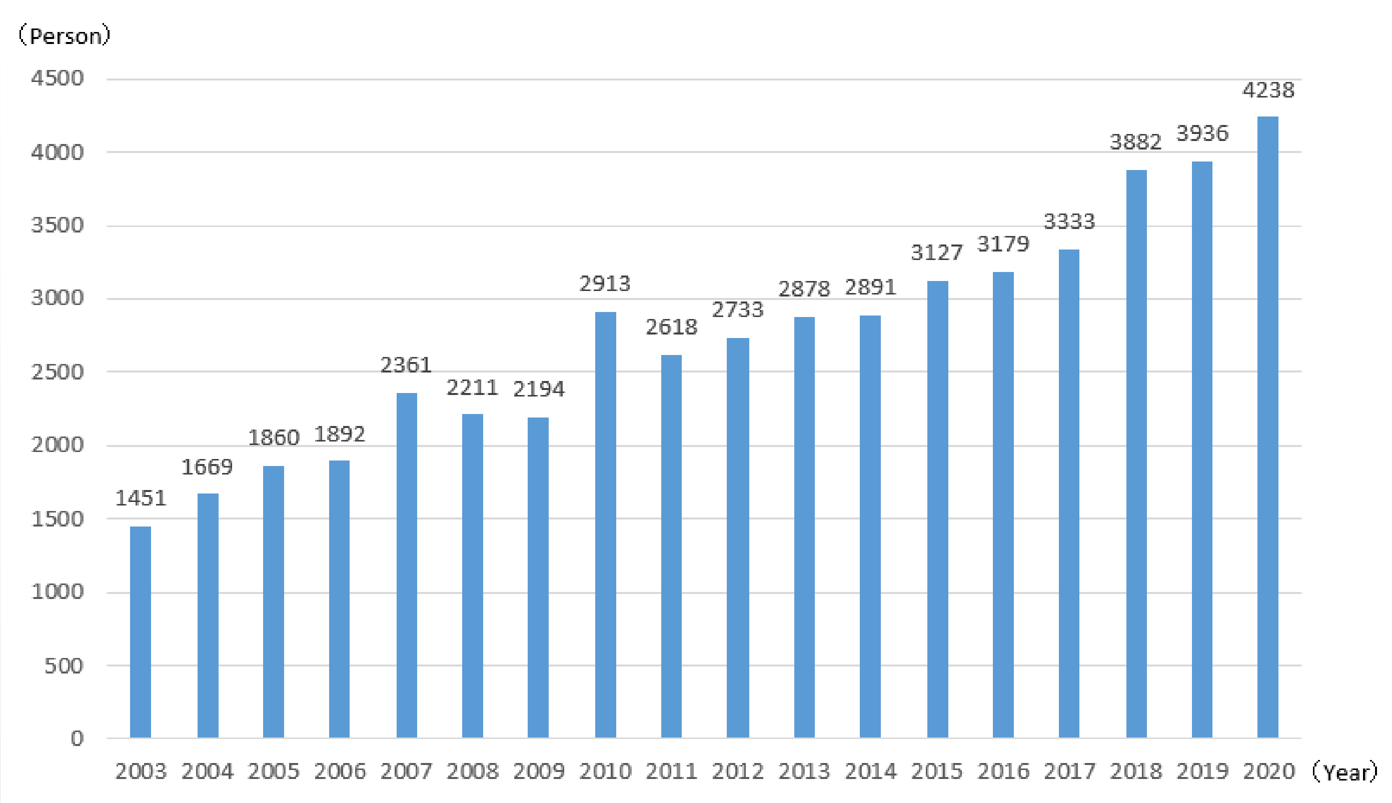

(From the 2018 Cabinet Office White Paper on Aging Society)

Figure 2 Changes in the proportion of single-person households aged 65 and over

In Japan, the number of single-person households is on the rise. Among them, the percentage of single-person households aged 65 and over will be about 40% in 2020, for both men and women.

# Social issues brought about by the aging of Japan

## 【Issue 1】Increase in lonely death




(From the 2021 Cabinet Office White Paper on Aging Society)

**Figure 3 Number of deaths at home for people aged 65 and over living alone in the 23 wards of Tokyo**

The number of deaths at home among people aged 65 and over living alone in the 23 wards of Tokyo was 4,238 in 2020, which is on the rise.

# 【Issue 1】Difference between average life expectancy and healthy life expectancy

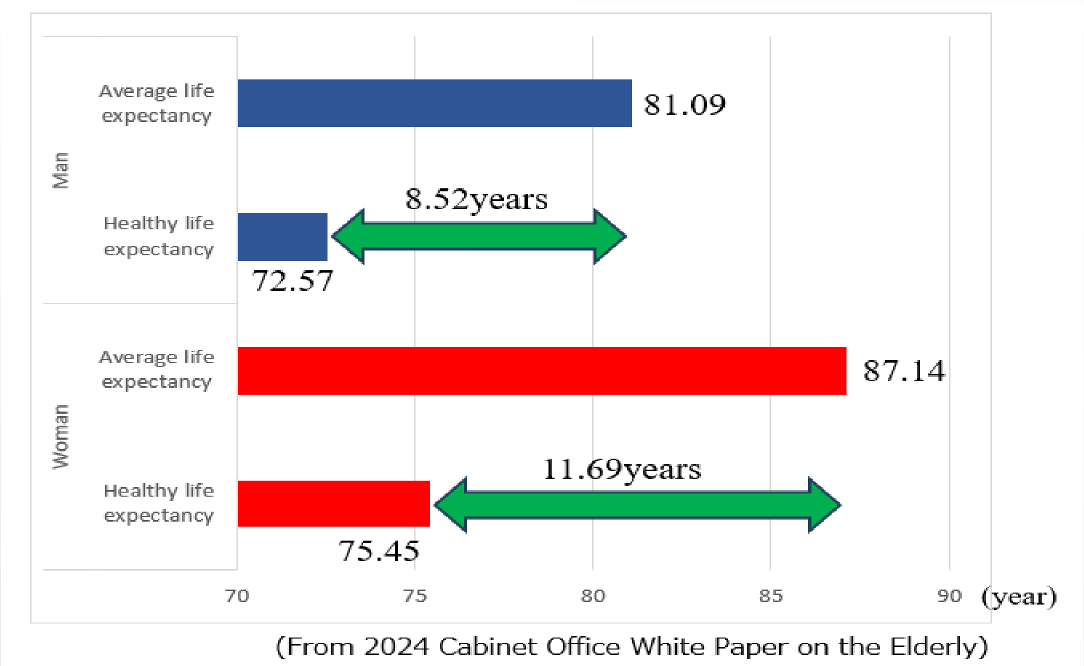



**Figure 4 Difference between average life expectancy and healthy life expectancy**

Average life expectancy> healthy life expectancy
➡ Increasing number of elderly people in need of treatment / care
➡ Pressure on medical expenses, pressure on bed utilization
➡ The country is for the elderly who need home care
Enhancement of long-term care insurance system that guarantees a.
Increase in shelter facilities
➡ Increase in social security spending

# Measures for Issue 1: Lively 100-year-old gymnastics

## What is 100 years old exercises Kumamoto prefecture version lively?

Kochi-shi made these exercises in 2002;, based on "100 years old exercises, get inhabitants representative and Local Elderly Care Management Center, cooperation including area rehabilitation wide area support center lively", and of Kumamoto original was made as 100 years old exercises lively. It is exercises used now at place of many commutes in the prefecture. It is warm-up, muscular strength campaign, exercises of about 40 minutes in total including rearranging exercises. By slow exercise, we can do it without unreasonableness at home. You will go 1-2 times a week.


Figure 5 Lively 100-year-old gymnastics Kumamoto version

As an initiative to prevent nursing care, prefectures are implementing "lively 100-year-old gymnastics".

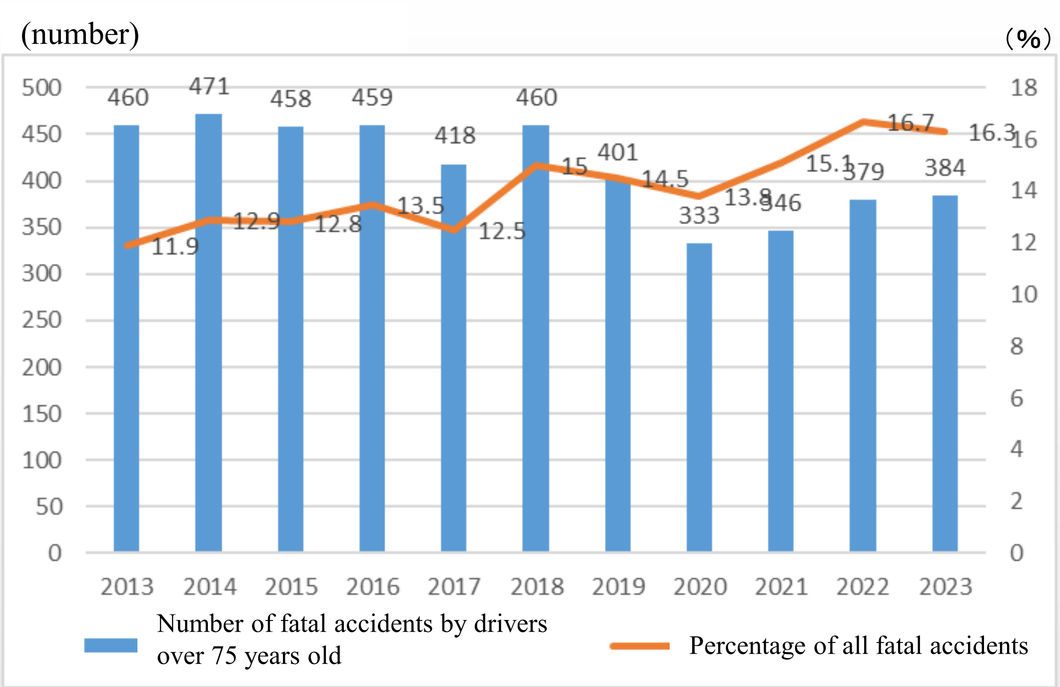
# Countermeasures for Issue 1: Watching service

Living away from older parents makes it difficult to notice when their parents suddenly feel sick or in the unlikely event that something happens. In such a case, the "watching service" is to watch over the parents on behalf of the family far away.

Elderly people wear sensors and press the button on the sensor to notify the registered company when something goes wrong, and the registered company visits the elderly people's homes to confirm their safety.

Figure 6 Elderly people wearing sensors

# [Issue 2] Increase in the number of fatal accidents caused by drive




(From the 2024 Cabinet Office "Traffic Safety White Paper")

**Figure 7 Changes in the number of fatal accidents caused by drivers aged 75 and over**

・Regarding fatal accidents caused by drivers aged 75 and over, the composition ratio to the total number of fatal accidents is on the rise, accounting for 16.3% of the total in 2023.

**Personal factors of traffic fatal accidents by drivers over 75 years old**

(1) Accident due to improper operation of handles, etc.
(2) Inattentive forward carelessness (involuntary driving, etc.)
(3) It occurs in the order of unconfirmed safety.

[Accident cases by elderly drivers]

In November 2016, a man in his 80s driving an ordinary passenger car made a runaway vehicle and sat on a bench when he stepped on the accelerator pedal by mistake for a brake pedal in a parking lot of a hospital in Shimotsuke City, Tochigi Prefecture. As a result of colliding with a woman and a pillar of a building, one woman died and two women were seriously injured.

Figure 8 Accident site

【Factors that drivers over 75 years old are more likely to have a fatal accident】

①It becomes difficult to obtain information on the surrounding situation due to weakened eyesight, etc., and the judgment becomes inadequate.
②Immediate response is delayed due to dull reflexes, etc.
③Due to the overall decline in physical strength, driving operations may become inaccurate.
④It may be difficult to continue driving for a long time.
⑤Driving becomes self-centered and it becomes difficult to objectively grasp the traffic environment.

# Technology development for Issue 2
## [Technology development for autonomous driving]

Autonomous driving is a millimeter-wave radar, which is a distance measuring sensor that has excellent weather resistance and measures distance, angle, and relative speed with high accuracy, and an ultrasonic sensor that detects vehicles entering the moving lane. Driving a car by detecting the position of the moving vehicle, detecting the driving state, detecting obstacles around the vehicle, etc. using a camera that recognizes images of the front and rear, GPS that detects position information, etc. It is a technology that controls automatically and safely.

Figure 9 EyeSight stereo camera

Figure 10 EyeSight recognition image

Let's take SUBARU's EyeSight as an example to explain autonomous driving. As shown in Figure 9, two cameras (stereo cameras) installed in the front of the vehicle monitor the front and three-dimensionally recognize obstacles to control automatic braking, cruise control, etc. I will. Figure 10 shows the image recognition image of EyeSight. The yellow frame recognizes vehicles, pedestrians, utility poles, center lines, etc. in front of you.

Even if an elderly person mistakenly steps on the accelerator and brake, the automatic brake will operate and accidents can be prevented.

# 【Issue 3】Cause of death peculiar to the elderly

| Ranking | 2017 | 2023 |
| --- | --- | --- |
| 1 | Malignant neoplasm(27.8%) | Malignant neoplasm(24.6%) |
| 2 | Heart disease(15.2%) | Heart disease(14.8%) |
| 3 | Pneumonia(8.7%) | Senility(11.4%) |
| 4 | Cerebrovascular disease(8.2%) | Cerebrovascular disease(6.9%) |
| 5 | Senility(7.2%) | Pneumonia(4.7%) |

(From the 2023 Ministry of Health, Labor and Welfare statistical data)

**Figure11 Leading causes of death for the elderly**

Increasing number of elderly people hospitalized for pneumonia
➡ Pressure on medical expenses, pressure on bed utilization
➡ Incorporated swallowing training into rehabilitation as an initiative to prevent "aspiration", which is the cause of pneumonia.

# Technology development for Issue 3【G O K U R I】

The University of Tsukuba / University of Tsukuba Hospital conducts research and development of the swallowing ability measuring instrument "GOKURI" with the cooperation of the government, companies, and medical institutions, and sells it as a venture company from the University of Tsukuba.
Food and saliva are sent from the oral cavity to the stomach via the pharynx and esophagus.
Accidental entry of food into the pharynx and trachea for some reason is called "aspiration."

Figure 12 "GOKURI"

Elderly people are more likely to have "aspiration" because their ability to swallow is reduced, and in Japan they develop aspiration pneumonia due to decreased swallowing function. By the way, pneumonia is the third leading cause of death in Japan.

"GOKURI" measures swallowing sound and posture with a sensor worn around the neck, and AI technology quantifies swallowing ability. This has made it possible to detect elderly people who are prone to aspiration at an early stage and start interventions such as swallowing exercises at an early stage.

# Data used in this course

**【System environment to acquire】**

Equipment: Thermometer, sphygmomanometer, activity meter, body composition meter

Communication method: Bluetooth

Data collection method: Data measured at home is automatically sent to the center using a smartphone, etc.

**【Target】** 65 years old and over

**【Acquisition frequency】** Measured once a day (before breakfast)

**[Period]** 1 month

Fig. 13 Display of measuring equipment and measurement results

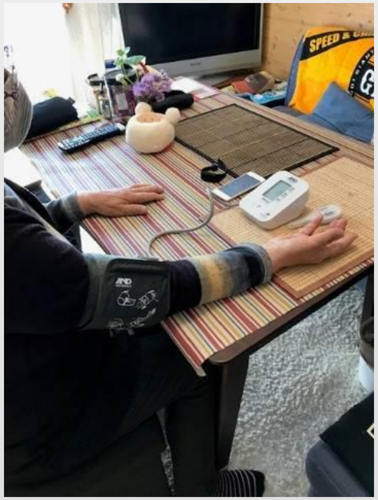


Fig. 14 Scene measuring at home

**【Acquired information】**

1. Vital related
  - ・Body temperature related file (A)
  - ・Blood pressure related file (A)
  - ・Weight related file(Body Fat Percentage, Basal Metabolism, Skeletal Muscle Percentage, Weight, BMI, Visceral Fat Level, Body Age) (A)
  - ・Calorie consumption related file(Step , Calorie , Sleep State , Sleep Hour, Sleep Minute) (A)
  - ・Home / indoor related files (A)

2. Subject attribute file (gender, age, medication, etc.) (B)
3. Inspection record file (B)
4. Life level related
  - ・Life independence file (B)
  - ・Cognitive function file (B)

5. Evaluation result file (C)
6. Meteorological files (C)

**【Evaluation】**

Evaluator: Specialist,

Evaluation date: One month after the start of measurement